Tempo — Project Guide & TA Viva Cheat Sheet

Plain-English Concepts, Slide Deck Outline, and Word-for-Word Speaking Scripts

Course: IKO 103 (Holistic Wellbeing) | Institution: IIT Roorkee | Target: Stress Detection & Triguna | Architecture: 100% On-Device & Privacy Preserving

Part 1 The 30-Second Elevator Pitch (What is Tempo?)

🎯 The Big Picture in Plain English

Tempo is an interactive web tool that estimates a person's real-time psychological stress using just their normal laptop webcam—**without any video ever leaving the browser or being recorded**.

In our course (**IKO 103: Holistic Wellbeing, Indic Perspective**), we explore how inner mental states reflect outwardly. Tempo connects this ancient Indic framework of **Triguna (Sattva, Rajas, Tamas)** with modern computer vision. Instead of judging everyone against an arbitrary average, Tempo first learns *your personal resting face*, and then detects how much your facial muscles, blinking, and movement deviate when you get stressed.

🎙️ **LINE TO SAY TO THE TA WHEN THEY ASK: "TELL ME BRIEFLY WHAT YOUR PROJECT DOES"**

"Tempo is an on-device wellness tool for IKO 103 that monitors psychological stress in real-time through the webcam. Its biggest feature is privacy: zero video or audio is ever recorded or uploaded—it all runs locally in the browser. It compares real-time facial micro-tensions against a calibrated personal baseline, and maps stress responses to the Indic framework of Triguna—Sattva for calm balance, Rajas for acute stress, and Tamas for fatigue."

Part 2 How It Actually Works (The 5 Core Ideas — No ML Jargon)

1 Privacy-First On-Device Computing (Why it's safe)

Most commercial AI uploads your camera feed to a cloud server. **Tempo never does that.** We use Google's lightweight MediaPipe vision model embedded inside the browser using WebAssembly. The browser processes camera frames on the CPU/GPU locally. The only data that can ever be exported is a CSV spreadsheet of anonymous numbers when the user explicitly clicks "Export".

2 The Invisible 3D Dot Grid (Facial Landmarks & Action Units)

The webcam doesn't look at colors or skin appearance. It tracks 468 invisible landmark points across the face. When facial muscles contract (for example, furrowing brows or tightening eyelids), the distances between these points change. In psychology, these micro-movements are called **FACS (Facial Action Coding System) Action Units**. Tempo specifically tracks:

- **AU4 (Brow Lowering):** Furrowing the eyebrows together (classic frown/concentration sign).
- **AU7 (Lid Tightening):** Squinting or tightening the lower eyelids when stressed.
- **Blink Rate:** How frequently you blink per minute.
- **Lip Tightening & Smile:** Tension around the mouth or relaxing smiles.

3 The "Resting Face" Problem & Personal Baseline (Z-Scores)

The Problem: Some people naturally have hooded eyes, low brows, or a serious resting face. If you used a universal threshold, a naturally serious person would always look "stressed."
Our Solution: Every session starts with an **80-second setup** (20s settling in + 60s quiet baseline calibration). Tempo calculates your personal average (mean μ) and natural variation (standard deviation σ) for each muscle. During the study, it calculates a **Z-Score**: *How many units are you tensing compared to your OWN normal?* A score of $+2.0$ means your eyelids are twice as tight as your usual resting state.

4 The Tension Formula (Why Blinking is Negative!)

Tempo combines these normalized muscles into a single **Facial Tension Index**. The formula gives heavy positive weight to eye tightening ($+1.0$) and brow lowering ($+0.5$).
Surprising Insight: Blink rate has a *negative weight* (-0.4). Why? When humans experience high mental workload or cognitive stress, their blinking actually **decreases** because they stare intently at the screen. Low blink rate + tight eyelids = cognitive strain!

5 Connecting to Indic Wisdom: The Triguna Framework

In Samkhya philosophy and Ayurveda, human psychological states are governed by three primary qualities (Gunas):

Sattva — Balance & Clarity

Relaxed, centered, peaceful state. Smooth breathing, stable gaze, open resting facial muscles, low tension.

Rajas — Restless & Agitated

High arousal, acute mental stress, fight-or-flight response. Furrowed brow, tight eyelids, elevated tension, fidgeting.

Tamas — Dullness & Sluggishness

Mental lethargy, low energy, fatigue. Drooping eyelids (high PERCLOS), low head movement, slumped posture.

Part 3 The 3-Phase Experimental Protocol (How to Test Participants)

To validate whether Tempo actually detects stress, we designed a rigorous 3-phase scientific protocol (minimum 90 seconds per phase):

PHASE	WHAT PARTICIPANT DOES	BUILT-IN INTERACTIVE TOOL	EXPECTED BIO-SIGNAL
1. Baseline (90s)	Sits quietly and breathes normally facing the screen.	Automated 80s calibration (mean & variance stored).	Tension Index near 0.0 (Z-score resting baseline).
2. Stressor (90s)	Faces acute cognitive conflict under time pressure.	**Built-in Stroop Test:** 7-color cognitive interference (e.g. word "BLUE" printed in RED ink).	Tension spikes ($>+1.5$), brow furrows, blink rate drops (Rajas state).
3. Recovery (90s)	Calms down the nervous system using guided breathwork.	**Built-in Breathing Coach:** Animated orb with 4-7-8 relaxing or 5-5 coherent breathing.	Tension returns toward baseline or lower (Sattva state).

> ⚠️ **Key Protocol Rule to Remember**
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> At the end of each phase, the participant completes a 4-slider self-report (Stress, Sattva, Rajas, Tamas from 0 to 10). **Rule:** The report must be saved *before* changing the phase switch, because the labels apply retrospectively to the preceding 60 seconds.

Part 4 Slide-by-Slide Outline for Your Presentation Deck

Use this exact 7-slide structure for your presentation. It tells a clear, cohesive story from the Indic problem statement to engineering and results.

Slide 1 Title & Motivation

- **Title:** Tempo: Real-Time On-Device Stress & Triguna Estimation
- **Context:** IKO 103 — Holistic Wellbeing (Indic Perspective), IIT Roorkee
- **Core Problem:** Stress is subjective, yet traditional biometric equipment (ECG, EEG) is expensive and invasive. Can we use ubiquitous laptop webcams privately?

Slide 2 Indic Framework: The Trigunas

- Ayurvedic & Samkhya concept of mind-body balance.
- **Sattva:** Equilibrium, peaceful focus $\rightarrow$ low muscle tension.
- **Rajas:** Hyper-activation, acute stress $\rightarrow$ facial tension, suppressed blinks.
- **Tamas:** Inertia, fatigue $\rightarrow$ slow eye closure, head slump.

Slide 3 Privacy-First Architecture

- **Video Never Leaves Browser:** 100% on-device computation via WebAssembly & MediaPipe.
- No cloud backend, zero video storage or recording.
- Meets strict privacy requirements for educational/psychological research.

Slide 4 The Personal Baseline (Z-Scores)

- Why universal thresholds fail (every face has a different resting geometry).
- 80s initial calibration to capture personal mean and standard deviation.
- All live measurements are computed as Z-scores ($z = (x - \mu)/\sigma$).

Slide 5 Biomechanical Face Mesh & Tension Index

- FACS Action Units: Brow furrow (AU4), eyelid tightening (AU7).
- Blink suppression during cognitive load.
- Downward head-pitch compensation: prevents false stress spikes when looking down at a desk or notebook.

Slide 6 Study Protocol & Interactive Tools

- **3-Phase Design:** Baseline $\rightarrow$ Stressor $\rightarrow$ Recovery (90s each).
- **Built-in Stroop Test:** 7-color cognitive interference stressor.
- **Built-in Breathing Coach:** 4-7-8 and 5-5 coherent breathing for parasympathetic vagal stimulation.

Slide 7 Results, Honest Limitations & Future Work

- **Results:** Clear directional divergence between Baseline and Stressor phases in automated tests.
- **Honest Academic Limitations:** Facial tension weights are literature priors, not yet clinically validated; remote pulse (rPPG) works on synthetic data but is sensitive to ambient lighting on real webcams.
- **Future Work:** Collecting diverse multi-subject study data across varied lighting conditions and evaluating against phase-mean baselines.

Part 5 TA Viva Cheatsheet — Exact Scripts to Say Out Loud

Here are direct, authentic answers to the exact questions TAs typically ask during viva or evaluation. Memorize or keep these key phrases in mind!

? TA ASKS: "DID YOU TRAIN A DEEP LEARNING MODEL YOURSELF, OR WHAT MACHINE LEARNING IS HAPPENING HERE?"

"We use Google's pre-trained MediaPipe Vision model to extract 3D facial landmarks and action-unit blendshapes directly in the browser. On top of that, our system performs real-time statistical normalization: it calibrates the user's personal mean and standard deviation to compute continuous Z-scores. We also designed a machine learning pipeline in Python (`train_guna.py`) that evaluates supervised Ridge and Logistic models trained on retrospective self-reports against both constant and phase-mean baselines."

TA impression: Shows you understand both the real-time feature extraction pipeline and the training evaluation architecture.

? TA ASKS: "WHY DO YOU HAVE AN 80-SECOND CALIBRATION PERIOD BEFORE STARTING?"

"Because human faces have diverse neutral anatomy. If someone naturally has low eyebrows or a narrower eye aperture, a fixed threshold would falsely classify them as permanently stressed. The 80-second calibration—20 seconds of settling in and 60 seconds of quiet baseline—measures each person's unique resting mean and variance. This ensures we only measure relative change against their own baseline."

TA impression: Proves you understand biomechanical variability and experimental control.

? TA ASKS: "HOW DID YOU CONNECT MODERN COMPUTER VISION TO THE INDIC TRIGUNA CONCEPT?"

"In Indic psychology from Samkhya and Ayurveda, internal mental states reflect in the physical sheath (Annamaya and Pranamaya koshas). We mapped physiological and facial activation to the three Gunas: Rajas corresponds to sympathetic nervous system activation—facial tension, brow furrowing, and restless fidgeting. Sattva corresponds to parasympathetic balance—relaxed facial musculature and steady gaze. Tamas corresponds to fatigue and sluggishness—characterized by high eyelid closure duration (PERCLOS) and postural slump."

TA impression: Perfectly links the technical project to the IKO 103 course curriculum!

? TA ASKS: "IS THIS CAMERA FEED RECORDING THE USER OR UPLOADING PHOTOS TO A SERVER?"

"No, absolutely not. Privacy is a non-negotiable invariant of our system. The video stream is processed in real-time memory frame-by-frame inside the browser client via WebAssembly. No video, image, or audio is ever recorded, stored on disk, or sent over the network. The only data that leaves the system is anonymous numerical metrics when the user manually clicks Export CSV."

TA impression: Demonstrates strong ethical awareness and user privacy compliance.

? TA ASKS: "WHAT ARE THE LIMITATIONS OF YOUR PROJECT? WHAT DIDN'T WORK?"

"We want to be scientifically honest about our current limitations: First, our facial tension index weights are literature-informed priors rather than clinically validated weights. Second, while our facial action units are robust, our optical pulse detection (rPPG) passed synthetic unit tests but remains noisy under varying real-world room lighting. Third, we also identified a foreshortening artifact where tilting the head down mimics brow tension, so we implemented a downward-pitch compensation filter."

TA impression: Saying plainly what is unvalidated or experimental is the #1 hallmark of top researchers and always earns high marks.

🤝 **HOW TO DIVIDE THE WORK SMOOTHLY WHEN SPEAKING:**

"My partner focused primarily on the low-level signal processing, WebAssembly packaging, and testing pipeline. My focus was on the human-computer interaction, the Triguna theoretical framework, the experimental 3-phase protocol design, and crafting our presentation."

💡 *TA impression: Demonstrates clear collaboration, mutual respect, and legitimate individual contribution.*

Part 6 Quick-Glance Feature Reference

SIGNAL / FEATURE	WHAT IT MEASURES IN PLAIN ENGLISH	WHEN STRESSED (RAJAS)	WHEN CALM (SATTVA)
AU4 (Brow Lowering)	Eyebrows drawn together and down (frowning/concentration)	High positive Z-score	At resting baseline (0.0)
AU7 (Lid Tightening)	Lower eyelids pulled upward and squinted	High positive Z-score ($+1.0$ weight)	Relaxed, soft eyelids
Blink Rate	Number of blinks per minute	**Drops significantly** (suppressed by intense focus)	Normal spontaneous rate (12–18/min)
AU12 (Smile)	Corners of the mouth pulled up by zygomatic muscle	Negative tension weight (relieves stress)	Soft, neutral or gentle smile
Head & Body Motion	Sway, shoulder shrugs, and postural micro-movements	Fidgeting or stiff freezing	Smooth, steady, centered posture